Article

Molecular architecture of the developing mouse brain

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Gioele La Manno[1,2,5]✉, Kimberly Siletti[1,5], Alessandro Furlan[1,4], Daniel Gyllborg[3], Elin Vinsland[1], Alejandro Mossi Albiach[1], Christoffer Mattsson Langseth[3], Irina Khven[2], Alex R. Lederer[2], Lisa M. Dratva[2], Anna Johnsson[1], Mats Nilsson[3], Peter Lönnerberg[1] & Sten Linnarsson[1]✉

The mammalian brain develops through a complex interplay of spatial cues generated by diffusible morphogens, cell–cell interactions and intrinsic genetic programs that result in probably more than a thousand distinct cell types. A complete understanding of this process requires a systematic characterization of cell states over the entire spatiotemporal range of brain development. The ability of single-cell RNA sequencing and spatial transcriptomics to reveal the molecular heterogeneity of complex tissues has therefore been particularly powerful in the nervous system. Previous studies have explored development in specific brain regions[1–8], the whole adult brain[9] and even entire embryos[10]. Here we report a comprehensive single-cell transcriptomic atlas of the embryonic mouse brain between gastrulation and birth. We identified almost eight hundred cellular states that describe a developmental program for the functional elements of the brain and its enclosing membranes, including the early neuroepithelium, region-specific secondary organizers, and both neurogenic and gliogenic progenitors. We also used in situ mRNA sequencing to map the spatial expression patterns of key developmental genes. Integrating the in situ data with our single-cell clusters revealed the precise spatial organization of neural progenitors during the patterning of the nervous system.

We used droplet-based single-cell RNA sequencing to profile the embryonic mouse brain each day between embryonic day (E)7 and E18 (Extended Data Fig. 1a, b, Supplementary Table 1). After removing low-quality cells and doublets (Methods), 93 samples and 292,495 cells remained, with a mean of 5,766 transcripts (unique molecular identifiers (UMIs)) and 1,934 genes detected per cell (Extended Data Fig. 1c–f). Total cellular RNA content decreased with embryonic age in all lineages and then increased again in the most mature neurons (Extended Data Fig. 1e, f, Extended Data Fig. 2c, d). Erythrocytes segregated into primitive and definitive erythropoiesis and showed exceptionally high UMI counts, but low gene counts (Extended Data Fig. 2a). A transcriptional signature of cell-cycle activation was bimodal, with 32% of all cells cycling (Fig. 1, Extended Data Fig. 2b). Postmitotic cells expressed a larger proportion of nascent (unspliced) RNA relative to mature (spliced) RNA (Extended Data Fig. 2c–e). Spliceosome genes showed the opposite pattern, suggesting a general mechanism for the regulation of splicing rates during brain development (Extended Data Fig. 2f).

Using an updated version of the Cytograph pipeline (Methods, Extended Data Fig. 3), we found 798 clusters (Fig. 1). We performed a sensitivity analysis, which confirmed robustness against different meta-parameters but highlighted that the stopping condition determines the number of clusters obtained (Extended Data Fig. 2g–i). We manually annotated and curated clusters with respect to putative major class (for example, fibroblast, vascular, immune, radial glia, glioblast, neuroblast and neuron), subclass, cell identity and spatial location (Supplementary Table 2).

A *t*-distributed stochastic neighbour embedding of the dataset (Fig. 1) revealed that cells were organized primarily by gestational age, and branched to show the major neuroepithelial lineages. A large connected component—containing the neural tube and its derivatives—was surrounded by disconnected islets of microglia, erythrocytes and vascular cells that do not derive from the neuroepithelium. At E7, cells segregated into the three germ layers generated during gastrulation. The neural crest formed during E8–E10 and differentiated during E10–E16 into the fibroblast lineage that formed the meninges[11]. The early neural tube formed at E8 and then matured into proliferating radial glia, the precursors of all neural cells. Two large cohorts of neurons branched from the neural tube, one comprising forebrain excitatory neurons and one containing all other neuronal types. The clear separation of forebrain excitatory neurons agrees with previous findings in the adult mouse brain[8], indicating the strong transcriptional and functional specialization of those cell types. After about E14, radial glia gradually lost proliferative capacity and switched to a glioblast state that eventually generated astrocyte, ependymal and oligodendrocyte precursor (OPC) cells. Major classes of cells appeared in successive waves that

[1]Division of Molecular Neurobiology, Department of Medical Biochemistry and Biophysics, Karolinska Institute, Stockholm, Sweden. [2]Brain Mind Institute, School of Life Sciences, École Polytechnique Fédérale de Lausanne, Lausanne, Switzerland. [3]Science for Life Laboratory, Department of Biochemistry and Biophysics, Stockholm University, Solna, Sweden. [4]Present address: Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, USA. [5]These authors contributed equally: Gioele La Manno, Kimberly Siletti. ✉e-mail: gioele.lamanno@epfl.ch; sten.linnarsson@ki.se

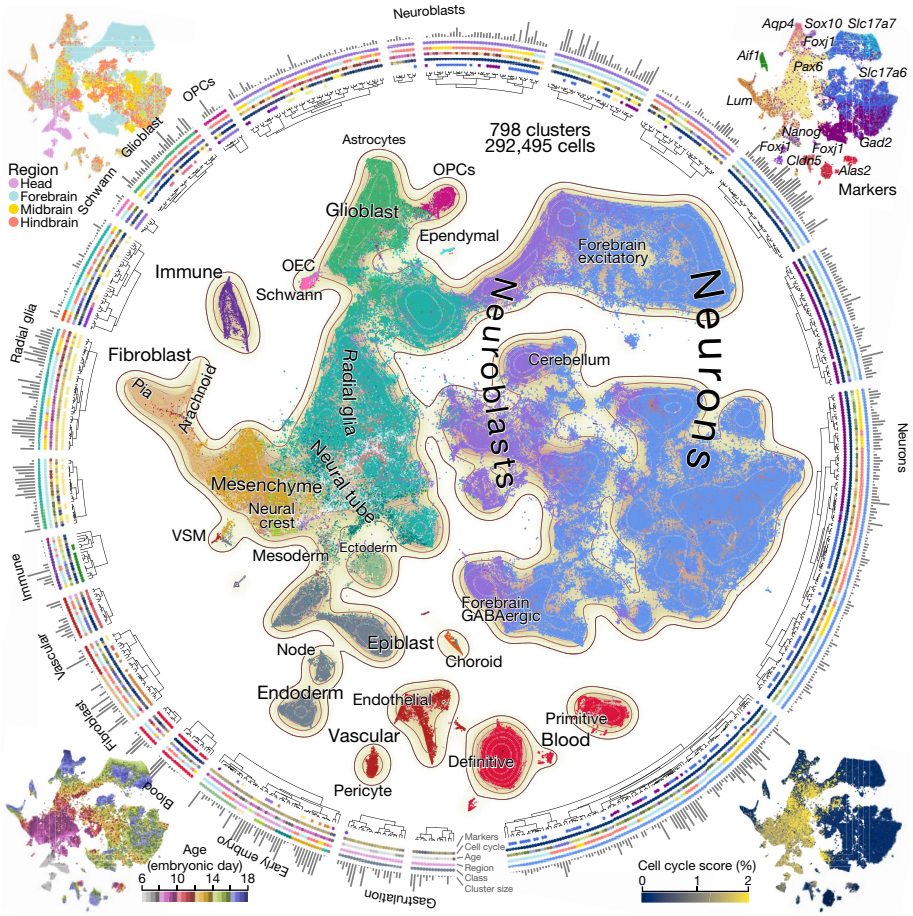

Fig. 1 | Atlas of the developing mouse brain. A *t*-distributed stochastic neighbour embedding (*t*-SNE) (centre) shows 292,495 cells from 93 pooled samples of 215 embryos, coloured by major classes. Clockwise from the top left, the four corner insets show the dissected region, marker-gene expression, cell-cycle score and pseudoage (average gestational age of a cell and its 25 nearest neighbours). The surrounding radial dendrogram shows 798 clusters divided into 25 categories, as indicated by gaps. Grey bars indicate the number of cells in each cluster.

revealed the coordinated temporal development of the mammalian brain (Extended Data Fig. 2j).

Focusing first on late gastrulation and early neurulation, we extracted E7–E8.5 cells from our dataset (Extended Data Fig. 4a, b) and divided the population into five parts according to the cluster dendrogram (Extended Data Fig. 4b, c). These cells included primitive-streak-stage (*Pou5f1*[12] and *T*), non-neural lineage-restricted (*Foxa2* and *Krt8*[13,14]) and early neuroepithelial (*Pax6* and *Hes3*[15,16]) progenitors. Moreover, our data aligned to a previously described single-cell atlas of gastrulation[17] (Extended Data Fig. 4d–l). We separated and further analysed our neuroepithelial clusters (*Fez1*, *Sox3*, *Pax6* and *Pdzph1*[18]) (Extended Data Fig. 4m–o) and distinguished ten subpopulations that differentially expressed genes linked to spatial patterning[19,20] (Extended Data Fig. 4n), including factors that pattern the forebrain and eye fields (clusters designated NepitA1-2); midbrain, hindbrain, and spinal cord (clusters designated NEpitMd, NEpitMH, NepitH1-2 and NEpitC); and the floor plate (clusters designated NEpitVL). We also found small numbers of E7–E8.5 cells among our neuroblast clusters (Extended Data Fig. 4p–r). These clusters otherwise contained cells sampled mainly from the midbrain and hindbrain, which suggests that the first neuroblasts appear posteriorly. Two of the five clusters were probably cranial sensory neuroblasts (expressing *Tlx2* and *Six1*[21]), indicating early maturation of the sensory modalities.

During gastrulation, the anteroposterior axis of the neural plate is established by extrinsic signals. By contrast, brain morphogenesis after neurulation is orchestrated by intrinsic signals that emanate from specialized neural progenitors known as secondary organizers. We identified secondary organizers among our E9–E11 radial glia-like clusters by their expression of known morphogens, including *Shh*, and Wnt-, Bmp-, Fgf-, neuregulin- and R-spondin-family genes (Fig. 2a–d, Extended Data Fig. 5a, b). Organizer subtypes differentially expressed a large number of transcription factors, secreted ligands and surface receptors (Extended Data Fig. 5c, d). For example, *Wnt7b* and *Rspo2* distinguished dorsalizing organizers in the cortical hem, whereas *Wnt3* and *Fzd10* were specific to roof-plate organizers. *Ntf3* and *Sost* were expressed by mesencephalic and rhombencephalic roof-plate cells, respectively. All dorsalizing organizers expressed *Wnt9a*, *Wnt3a*, *Rspo1* and *Bmp6*. Although defined by *Shh* expression, each of the ventralizing organizers was further distinguished by region-specific expression of secreted molecules. For example, the hypothalamic floor plate was the only ventralizing organizer that expressed *Dlk1*, a Notch ligand, and was distinguished by the absence of *Spon1* and *Ntn1*. These results establish a rich repertoire of secondary organizers that secrete combinations of morphogens that might act in a region-specific manner to induce locally distinct neuronal fates.

A distinct clade of the cluster dendrogram comprised glia and related progenitors (Fig. 3a, Methods, Extended Data Fig. 6), and revealed two major groups: cells expressing neurogenic markers such as *Neurog2* or *Dlx2*, and those expressing glial markers such as *Tnc* or *Egfr* (Fig. 3b, c). We refer to the latter as glioblasts, given their likely commitment to gliogenesis. We found that more than 7,000 genes were differentially expressed across the two groups (Extended Data Fig. 6b, c), and glioblasts preferentially expressed astrocyte- and oligodendrocyte-related genes (Fig. 3d).

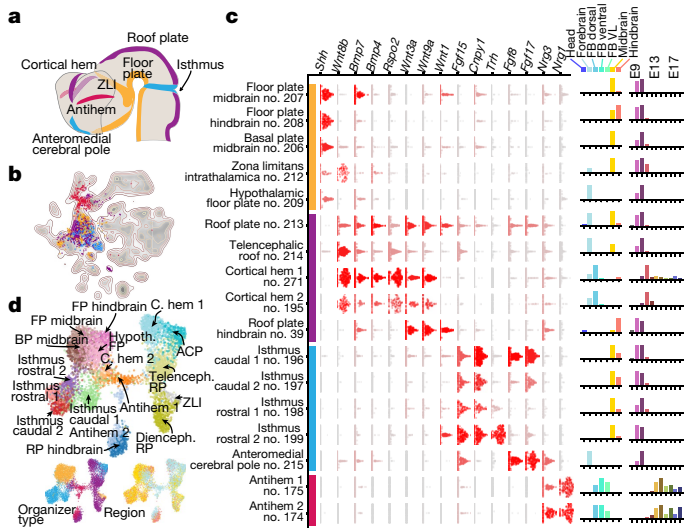

Fig. 2 | Molecular diversity of neural tube organizers. a, A schematic representation of secondary organizers in the neural tube. Colours group early organizers by ventralizing (yellow), dorsalizing (violet) and boundary-inducing (cyan) activity. Late organizers are fuchsia. ZLI, zona limitans intrathalamica. **b**, Cells belonging to organizer-related clusters coloured on the global *t*-SNE (Fig. 1). **c**, Beeswarm plots showing log-transformed UMI counts for morphogens that are known to define organizers. FB, forebrain; VL, ventrolateral. **d**, Uniform manifold approximation and projection embedding of organizer-related radial-glia-like cells. Miniatures are coloured by organizer and region. ACP, anteromedial cerebral pole; BP, basal plate; C. hem, cortical hem; dienceph., diencephalic; FP, floor plate; hypoth., hypothalamic; telenceph., telencephalic; RP, roof plate.

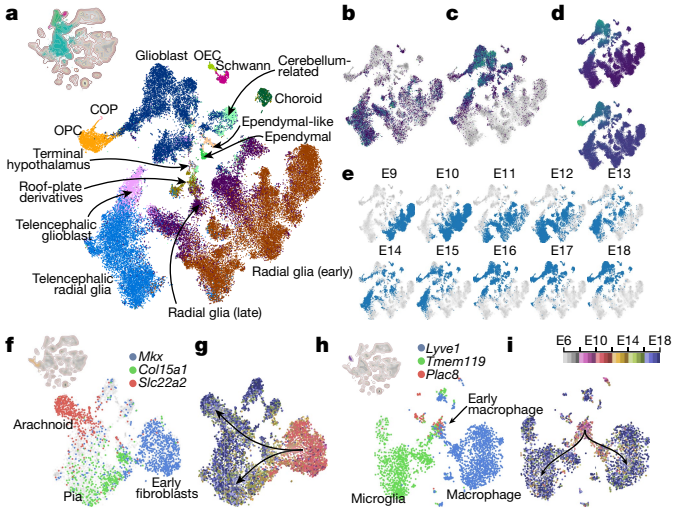

Fig. 3 | Non-neuronal cell types. a, Clusters in a clade of the dendrogram representing radial glia and related cells, analysed separately (inset shows included cells). The dendrogram was cut into 15 metaclusters, annotated on the *t*-SNE. COP, committed oligodendrocyte precursor; OEC, olfactory ensheathing cell. **b**, Cells coloured by the sum of pro-neural *Neurog2* and *Dlx2* expression. **c**, Cells coloured by the sum of *Tnc* and *Egfr* expression. **d**, Cells coloured by their astrocyte (top) or OPC (bottom) scores. **e**, Cells from the indicated gestational age are coloured on each *t*-SNE plot. **f**, Clusters in a clade of the dendrogram that contained fibroblasts, analysed separately (inset shows included cells). Colours indicate expression of *Mkx*, *Col15a1* and *Slc22a2*. **g**, Cells coloured by gestational age. **h**, Clusters in a clade of the dendrogram that contained immune cells, analysed separately (inset shows included cells). Colours indicate expression of *Lyve1*, *Tmem119* and *Plac8*. **i**, Cells coloured by gestational age.

Seven region-specific types of astrocyte in the adolescent mouse have previously been reported[8], including distinct telencephalic and non-telencephalic subtypes. In agreement, here we found that embryonic non-telencephalic glioblasts and OPCs of mixed forebrain, midbrain and hindbrain origin were distinct from telencephalic glioblasts (Fig. 3a; Extended Data Fig. 6d, e). These observations suggest a loss of regionalization, particularly in the oligodendrocyte lineage. To investigate further, we trained a classifier on each of six groups of cells (early radial glia, late radial glia, glioblasts, OPCs, neuroblasts and neurons) to identify the region of origin of a cell using transcription-factor expression (Extended Data Fig. 6f–h). We tested each classifier on the remaining groups; the classifiers performed well on radial glia and neuroblasts, but poorly on OPCs. Nevertheless, the classifier trained on glioblasts and OPCs performed reasonably well, probably relying on region-defining genes that were still present at low levels (Extended Data Fig. 6i).

The neurogenic-to-gliogenic transition occurred between E12 and E16 (Fig. 3e, Extended Data Fig. 6j). The earliest gliogenic cells appeared in a cluster that was unique in its co-expression of *Egfr* and the Notch ligands *Dll1*, *Dll3* and *Dll4* (Extended Data Fig. 6j, k), This cluster, which was closely related to OPCs and might represent a pre-OPC state, expressed high levels of *Olig1* and *Olig2*. The earliest astrocytes (*Gfap*, *Agt* and *Aqp4*) appeared at approximately E15 (Extended Data Fig. 6j).

Developing mesenchymal cells form a sheath known as the meninx primitiva that differentiates into the three meningeal layers: pia, arachnoid and dura mater[22]. Extending a previous single-cell study that characterized meningeal diversity at E14.5[23], we identified a branching meninges lineage that emanated from the early neural crest and mesenchyme (Fig. 1). Our detailed examination of these cells (Fig. 3f, g, Extended Data Fig. 7) revealed primitive fibroblast-like cell states that expressed *Mkx*, *Alx1*, *Alx3* and *Tbx18* at E9, which differentiated into pia (*Col15a1*) and arachnoid (*Slc22a2*) by E15. We also found dura (*Ccn3*, previously known as *Nov*) and perivascular fibroblast-like cells (*Tcf21*), but their relationship to the meninges lineage was unclear. We validated the meninges lineage using multiplexed single-molecule fluorescent in situ hybridization (Extended Data Fig. 7f, g). In agreement with the single-cell data, a single pool of cells expressed *Tbx18*, *Mkx*, *Alx1* and *Alx3* at E9.5; at E13.5 and E18.5, separate layers expressed *Col15a1* and *Slc22a2*.

Tissue-resident macrophages and microglia develop from the extraembryonic yolk sac and invade the brain before E9.5. We observed proliferating macrophages as early as E8 that persisted until E14 and were progressively replaced by microglia and perivascular macrophages (Fig. 3h, i, Extended Data Fig. 8). These early macrophages expressed distinct markers such as *Ybx3* and *Plac8*, as well microglia (*Tmem119*) and border macrophage (*Lyve1*) markers, which suggests that a superimposed progenitor state resolves into one of two opposing fates. High expression of *Spp1* denoted a subset of putative axon-tract-associated microglia[24].

To investigate gene expression along putative lineage trajectories, we inferred a pseudolineage tree for the dataset (Extended Data Fig. 9). We isolated all trajectories that ended in specific clusters and projected the single cells in each trajectory onto their embryonal pseudoages (Extended Data Fig. 9e–k). We then averaged gene expression in each pseudoage bin. To validate this approach, we examined the subtree terminating in cortical excitatory neurons (Extended Data Fig. 9h), the development of which is well-understood. As expected, only forebrain cells contributed to the trajectory. Furthermore, a set of transcription factors that regulate cortical excitatory neurogenesis was expressed in an ordered progression from E11 to E18: *Emx1*, *Eomes*, *Tbr1* and *Satb2*[5]. RNA in situ hybridization (Allen Brain Atlas[25]) confirmed the timing of this progression (Extended Data Fig. 9n).

Nearly all of the neuronal—but none of the glial—lineages passed through an early neuroblast state expressing all or most of the genes

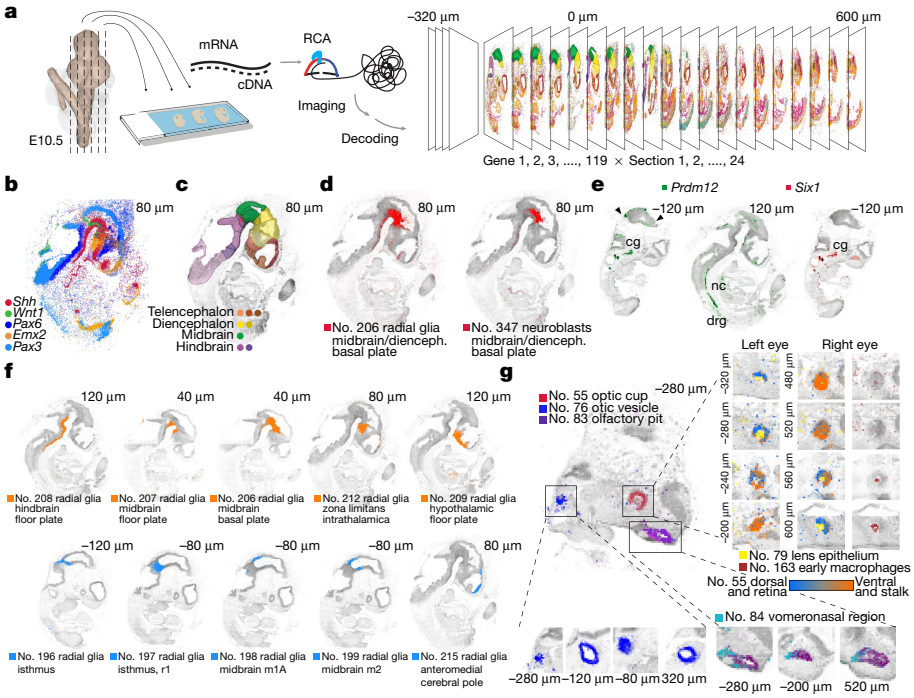

Fig. 4 | Spatial distributions of cell types and states. a, Overview of the in situ sequencing experiment, comprising 119 genes and 24 cryosections; performed twice with similar results. The distance of each section from the midline is indicated. RCA, rolling circle amplification. **b**, Example showing raw expression data for five genes on a single section. **c**, **d**, Manually annotated anatomical regions (**c**) and inferred distributions (**d**) of clusters no. 206 and no. 347 on the same section as **b**. **e**, Imputed *Prdm12* and *Six1* expression. cg, cranial ganglia; nc, notochord; drg, dorsal root ganglia; arrowheads bracket *Prdm12* expression in the dorsal midbrain and diencephalon. **f**, Inferred spatial distributions of ventralizing (orange) and boundary (blue) organizers. **g**, Clusters localized to the developing eye, ear and nose. Close-up images show spatial distributions over adjacent sections. Mouse 3D rendering in **a** was modified from EMAP eMouse Atlas Project (CC BY 3.0) (http://www.emouseatlas.org/).

Nhlh1, *Nhlh2*, *Ebf1*, *Ebf2* and *Ebf3*. The expression of *Nhlh1* and *Nhlh2* (Extended Data Fig. 9l) identified a brain-wide neuroblast state at or immediately after cell-cycle exit[26] (Extended Data Fig. 2b). Cells expressing these markers coincided with the set of clusters that we had manually annotated as neuroblasts, which formed a distinct clade in the cluster dendrogram (Fig. 1). Notably, forebrain GABAergic neuroblasts belonged to the same clade and expressed low levels of *Nhlh2*, but appeared to lack *Nhlh1* expression.

The developing brain comprises distinct spatial domains that express characteristic transcription factors. Over time, these domains are elaborated by regional interactions that involve diffusible morphogens and cell–cell contact. As a step towards linking this complex process to transcriptional cell states, we performed hybridization in situ sequencing[27] of 119 genes (Fig. 2c, Extended Data Fig. 5c, d) on 24 sagittal sections of an E10.5 mouse embryo (Theiler stage 17) (Fig. 4a, b, Extended Data Fig. 10a). We manually annotated anatomical regions using the Allen Developing Mouse Brain Reference Atlas ontology, guided by known gene-expression patterns and anatomical landmarks (Fig. 4c, Extended Data Fig. 10b). We used a generalized version of Tangram[28] (Methods) to align our single-cell clusters to the spatial images, yielding a probabilistic spatial profile for each cell state (Fig. 4d, Extended Data Fig. 10c–e). The trained Tangram model enabled us to impute spatial expression patterns for genes that were not directly measured by in situ sequencing, often with notable precision. For example, *Prdm12* and *Six1* were correctly localized in cranial and dorsal root ganglia and the notochord (Fig. 4e, Extended Data Fig. 10f); the imputation even correctly showed *Prdm12* but not *Six1* in the dorsal midbrain and diencephalon[21,29]. We used the spatial distributions to further refine the cluster annotations (Supplementary Table 2).

The floor, roof and boundary organizers that we identified by single-cell RNA sequencing (Fig. 2) were spatially segregated and expressed distinct sets of secreted ligands and receptors (Fig. 4f, Extended Data Fig. 10d). Our observations demonstrate that even progenitors born in adjacent spatial domains are subject to several distinct environmental cues. Although we selected genes for in situ sequencing to profile secondary organizers, we were able to map many non-neural clusters as well as more than 15 clusters related to developing cranial ganglia and sensory organs. For example, we mapped clusters to the optic cup, lens, otic vesicle and nasal pit. We could even detect residual spatial variability within clusters, including the transition between dorsally and ventrally patterned optic cup and stalk cells (Fig. 4g, Extended Data Fig. 10g). A subpopulation of early macrophages that mapped to the centre of the developing eye probably reflected hyalocytes, the resident macrophages of the vitreous[30]. This observation supports an extraembryonic yolk sac origin for hyalocytes (similar to microglia) rather than an origin in the bone marrow, which has not yet developed at this time point.

A central question in brain development concerns the origin of neuronal diversity. We observed that radial glia are spatially and transcriptionally diverse (Figs. 2, 4) and secrete a greater variety of diffusible morphogens than previously appreciated, providing precise local cues to specify neuronal cell types. At the same time, we observed only 87 radial glia subtypes, but 171 neuroblast and 306 neuron subtypes. The transition from radial glia to neuroblast was marked by several abrupt changes: cell-cycle exit (Extended Data Fig. 2b), a sudden increase in splicing activity (Extended Data Fig. 2e, f) and the expression of a transcription factor module unique to this transition and conserved across most lineages (Extended Data Fig. 9h–m). Overall, these observations support a model in which neural progenitors are broadly patterned by secondary organizers, but most neuronal diversity is generated postmitotically in neuroblasts and maturing neurons.

Our atlas provides an overview of the transcriptomic diversity of the prenatal brain, which is a step towards investigating the development of the mammalian nervous system. The wealth of information on

time-, lineage- and region-specific gene expression provides powerful tools for genetic targeting, and for understanding genes involved in neurodevelopmental disorders and brain cancer.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-021-03775-x.

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Methods

No statistical methods were used to predetermine sample size. The experiments were not randomized, and investigators were not blinded to allocation during experiments and outcome assessment.

Mice

CD-1 mice were obtained from Charles River and mated overnight. The morning that plugs were detected was considered E0.5. Mice were housed with a regular dark–light cycle and fed standard diet and water ad libitum. All mouse procedures were approved by the Stockholm ethics committee (N68/14; Stockholms djurförsöksetiska nämnd) and followed Directive 2010/63/EU of the European Parliament and of the Council, the Swedish Animal Welfare Act (Djurskyddslagen: SFS 1988:534), the Swedish Animal Welfare Ordinance (Djurskyddsförordningen: SFS 1988:539) and the provisions regarding the use of animals for scientific purposes (DFS 2004:15 and SJVFS 2012:26).

Dissections

Two hundred and fifteen embryos between E7 and E18.5 were collected from 45 pregnant mice. Embryos were extracted from the uterine horn and dissected in phosphate-buffered saline (PBS) on ice within one hour. Dissections were adapted to accommodate changes in size and cell numbers in the developing brain (Supplementary Table 1). E7 and E8 embryos were cut into two parts to isolate the rostroventral neural tube from dorsal tissue including the trophectoderm, which was discarded. The developing spinal cord was identified by corresponding somite pairs and also discarded. From E9 and E10 embryos, we isolated the entire cephalic part, including prospective forebrain, midbrain and hindbrain. From E11 onwards, only developing brain tissue was dissected and collected; the surrounding head tissue was discarded. Meninges were removed whenever possible, but could not be eliminated completely. E9 to E11 brains were divided into forebrain, midbrain and hindbrain. Forebrain and midbrain were separated at the diencephalon; midbrain and hindbrain at the isthmus. E12 to E15 forebrains were further divided into dorsal (including the hemispheres) and ventral (including the subcortical territories, caudoputamen and the ventral striatum) parts. E16 to E18 forebrains were divided into dorsal, ventral, and thalamic parts.

Cell dissociation

Dissected tissue was processed as previously described, with minor modifications[8]. Tissue was transferred to a 2-ml Eppendorf tube containing preheated digestion solution (37 °C). Before E11, trypsin was used for dissociation; after E11, the digestion solution contained 300 µl TrypLE Express (Life Technologies; cat. no. 12605-010), 2,000 µl papain solution (Worthington Biochemical; cat. no. LK003178; 25 U ml^{-1} in cutting solution), 100 µl DNase I (Worthington Biochemical; cat. no. LK003172; 1 mM in cutting solution) and 100 µl cutting solution. Enzymatic incubation time was adjusted on the basis of developmental stage. The tissue was kept in a 37-°C water bath and gently pipetted three times every 10 min. After 30 min at 37 °C, another 100 µl of TrypLE solution was added, followed by 10 more minutes at 37 °C and pipetting. After approximately 45–50 min, dissociation was complete, and cells became visible at the bottom of the Eppendorf tube. The solution was filtered using a 20-µm cell strainer (Falcon cat. no. 352340) and collected in a 15-ml plastic tube. The digestion solution was diluted with 2.4 ml of cutting solution and 0.6 ml of Neurobasal-A medium (Gibco; cat. no. 10888) and centrifuged at 100*g* for 4 min at 4 °C. The supernatant was removed and the pellet resuspended in 0.5 ml cutting solution and 0.5 ml Neurobasal-A medium. Neurobasal-A medium was supplemented with L-glutamine (Gibco; cat. no. 25030-123), B27 (Gibco; cat. no. 17504-044) and penicillin plus streptomycin (Sigma 50x; cat. no. P4458). The cell suspension was carefully transferred with a Pasteur pipette onto an Optiprep gradient. For the gradient, 85 µl of Optiprep Density Solution (Sigma; cat. no. D1556) was mixed with 457.5 µl cutting solution and 457.5 µl Neurobasal medium including supplements. The gradient was subsequently centrifuged at 80*g* for 10 min at 4 °C. The supernatant was removed until only 100–200 µl remained. DNase I (10 µl) was added to prevent aggregation. The single-cell density was evaluated. Then, cells were filtered with a 20-µm strainer (Partec CellTrics). Cells were pelleted, resuspended in cutting solution with DNaseI, and loaded into 10X Genomics Chromium v.1 chips.

Single-cell RNA sequencing

Droplet-based single-cell RNA sequencing was performed using the 10x Genomics Chromium Single Cell Kit v.1. Single-cell suspensions concentrated at 500–700 cells per ml were mixed with master mix and nuclease free water according to the Chromium manual, targeting 3,500 cells per reaction. Twelve PCR cycles were used for cDNA synthesis. Each library was sequenced using an Illumina HiSeq instrument, one sample per lane, with one 98-bp read located near the 3′ end of the mRNA. Illumina runs were demultiplexed and aligned to the genome (mm10-3.0.0), and BAM files were obtained from the 10X Genomics cellranger pipeline (version 3.0.2). Molecule counts were attributed to spliced and unspliced transcripts by running velocyto[31] (version 0.17.11) with standard parameters, resulting in one loom file (http://loompy.org) per sample.

In-situ sequencing by HybISS

HybISS[27] was performed as published at protocols.io[32]. Padlock probes, bridge probes and detection oligonucleotide sequences used are listed in Supplementary Table 3. Imaging was performed with a Zeiss Imager. Z2 epifluorescence microscope equipped with LED light source (Lumencor SPECTRA X), sCMOS camera (Hamamatsu ORCA-Flash4.0 LT+) and 20× objective (Zeiss Plan-Apo/0.8 air, 420650-9901). Regions of interest were imaged with 10% overlap and 24 *z*-stack planes with 0.5-µm spacing, maximum-projected in Zeiss Zen 3.2 software, and then tiles were exported as raw TIFF files.

Single-molecule RNA fluorescence in situ hybridization

Coverslips (Marienfeld) were rinsed twice with distilled water for 20 min and incubated for 24 h in concentrated nitric acid (Sigma Aldrich). Then, coverslips were washed with distilled water 4 times and stored in 95% ethanol. Before use, the slides were cleaned for 10 min with a plasma cleaner. After that, slides were immersed in 1% bind-silane (GE; 17-1330-01), 10% ethanol and 0.03% acetic acid solution for 1 h, rinsed 3 times in absolute ethanol and air-dried.

Fresh-frozen embryos at E9.5, E13.5 (whole) and E18.5 (head) were cut on a cryostat and 10-µm sections were collected onto the same functionalized coverslip. After mounting, the tissue sections were fixed for 10 min with PBS buffered 4% PFA (28908; Thermo Fisher), rinsed twice with PBS (AM9625, Ambion), dehydrated for 3 min in isopropanol, air dried and stored at −80 °C.

Encoding probes were designed composed of a target region between 26–32 bp and 4 equal 20-bp readout probe binding sites on the 3′ end. The target regions were designed to have a GC content between 40–60%, a Gibbs free energy at 37 °C below −25 kcal mol^{-1}. All candidate target regions were aligned against the mouse transcriptome and only those target regions with identity below 60% to off-target sequences were kept.

Tissue was rehydrated in 2× SSC (Sigma; S6639) and cleared three times 10 min with permeabilizing solution (8% SDS (Sigma; 05030) and 0.5% Triton X-100 (Sigma; T8787) in 200 mM boric acid pH 8.5). After that, encoding probes were hybridized (3 nM per probe) in 30% formamide hybridization buffer (2× SSC, 10% (w/v) dextran sulfate (Sigma; D8906), 30% (v/v) formamide (Ambion; AM9342), 1 mg ml^{-1} *Escherichia coli* tRNA (Roche; 10109550001), 2 mM ribonucleoside vanadyl complex (RVC) (Sigma; R3380) and 0.2 mg ml^{-1} BSA (Sigma-Aldrich) and 2 µM of each anchor probe (acrydite-polyT15 LNA and acrydite-polyT30

LNA)) for about 48 h at 38.5 °C. After encoding probe hybridization, tissue was washed 4 times 15 min with 30% formamide at 38.5 °C. Then, the tissue was rinsed with 4% acrylamide solution (Bio-rad; 1610144) in 2× SSC for 30 min at room temperature. To induce polymerization, the tissue was covered with 25 µl of 4% acrylamide solution, 2× SSC, 0.05% ammonium persulfate (Sigma; A3078S), 0.05% *N*,*N*,*N*′,*N*′-tetramethylethylenediamine (Sigma; T7024) and sandwiched with a GelSlick (Lonza; 50640) pretreated coverslip. Polymerization took place on a nitrogen chamber for 30 min at 4 °C and 2.5 h at 38.5 °C. After polymerization, coverslips were detached and the sample coverslip was placed on a flow cell and cleared with 5 ml of digestion solution (2× SSC, 1 mM EDTA (Ambion; AM9260G), 2% SDS, 0.25% Triton X-100, 0.08 U ml^{-1} Proteinase K (NEB; P8107S)) overnight at 37 °C. After clearing, the sample was hybridized with 50 nM of readout probes conjugated to either Atto 488, Cy3, Atto 590 or Alexa Fluor 647 in 10% formamide hybridization buffer (2× SSC, 10% (w/v) dextran sulfate 10% (v/v) formamide, 1 mg ml^{-1} tRNA, 2 mM RVC and 0.2 mg ml^{-1} BSA) for 1.5 h at 30 °C. Then, the sample was washed 3 times 10 min with 10% formamide and 1 mg ml^{-1} Hoechst (Sigma-Aldrich) at 30 °C, followed by 3 washes with 2× SSC. The sample was washed with anti-bleaching imaging buffer (2 mM Trolox (Sigma; 238813), 5 mM 3,4-dihydroxybenzoic acid (Sigma; 37580), 20 nM protocatechuate 3,4-dioxygenase (Sigma; P8279), 2× SSC) and imaging was performed at room temperature. After imaging, fluorescent dyes were cleaved by washing 3 times 10 min with 50 mM TCEP hydrochloride (Sigma; 646547) in 2× SSC at room temperature. After fluorophore cleavage, a second hybridization and imaging round was performed.

Imaging was performed on a standard epifluorescence microscope (Nikon Ti Eclipse, Nikon), equipped with a motorized stage (Nikon or HLD117 Prior), a Nikon CFI Plan Apo Lambda × 60 oil immersion objective with a numerical aperture of 1.4, Spectra X seven-line LED-light source (Lumencor) and Zyla 4.2 plus scMOS camera (Andor, 6.5 µm physical pixel size) for detection.

Computational analysis

Sample size, randomization and blinding. The total number of samples analysed was designed to ensure a minimum of three mice per tissue and time point (as indicated in Supplementary Table 1). As the study was descriptive and did not test a hypothesis or contrast, no randomization or blinding was performed. However, sex was automatically randomized owing to the fact that sex could not be controlled in these prenatal samples.

Preprocessing and preliminary clustering. Loom files from each 10x sample were aggregated by time point and brain region: cephalic neural tube E7–E8.5, forebrain E9–E11, midbrain E9–E11, hindbrain E9–E11, forebrain dorsal E12–E15, forebrain ventral E12–E15, midbrain E12–E15, hindbrain E12–E15, forebrain dorsal E16–E18, forebrain ventrothalamic E16–E18, forebrain ventrolateral E16–E18, midbrain E16–E18 and hindbrain E16–E18. Cells with fewer than 2,000 UMIs were excluded from pooling. Genes expressed in fewer than 10 cells or greater than 60% of cells were excluded from analysis.

We used the cytograph package (detailed in 'Cytograph 2.0') to recursively analyse each pool. Differential-expression testing was used to merge similar clusters. Doublets were removed by running a modified version of DoubletFinder (available in cytograph) on each sample in the dataset. DoubletFinder scores were averaged across clusters, and clusters with an average score greater than 0.3 were removed from further analysis. Approximately 20 additional clusters were removed that were manually annotated as blood-contaminated, doublets or containing fewer than five cells. The procedure retained 942 clusters.

Final clustering and dendrogram. A dendrogram of the 942 clusters was calculated and split into two, and each subset was processed again with cytograph. This procedure was repeated iteratively until all leaves contained 31 clusters or fewer, resulting in 798 clusters. Dendrograms were constructed using the Aggregator class of cytograph.

Cytograph 2.0

The analysis pipeline is available as an update to the Cytograph package, Cytograph 2.0 (https://github.com/linnarsson-lab/cytograph2). Cytograph is under continuous development, and the results in this paper were generated at commit 2e8eb79a5f83bd9dda15503d9b2476ebdd8fa0b3. Unless otherwise specified, the following parameters were used: k: 25, k_pooling: 10, factorization: HPF, nn_space: HPF, n_factors: 96, n_genes: 2000, mask: ('cellcycle', 'sex', 'ieg', 'mt'), remove_doublets: False, remove_low_quality: False, features: enrichment.

Cytograph v1.0 has previously been described[8], including gene enrichment (overexpression), trinarization and clustering. Cytograph uses the following open-source packages, and we are grateful to the authors for making these resources freely available to the community: numpy, scikit-learn, scipy, networkx, python-louvain, hdbscan, pyyaml, statsmodels, numpy-groupies, tqdm, umap-learn, torch, harmony-pytorch, pynndescent, click, leidenalg, unidip and opentsne. We detail below only the key improvements: changes to dimensionality reduction and graph construction; auto-annotation; and punchcards.

In brief, we used hierarchical Poisson factorization (HPF)[33] to decompose the expression matrix into 96 non-negative and modular components, while reducing noise and preserving most of the structural information in the original matrix. We then computed a radius nearest neighbour graph (RNN) using the information radius (also known as the Jensen–Shannon divergence (JSD), which is a measure of the Shannon information difference) to link cells with near-identical gene-expression states. The JSD measures the entropy of the average of two cells, compared to the average entropies of the individual cells. It is a symmetrized version of relative entropy (Kullback–Leibler divergence). For a pair of identical cells, their HPF component values are drawn from identical distributions, the entropy of the average is equal to the average of entropies and the JSD is zero. For a pair of dissimilar cells, the average of entropies is smaller than the entropy of the average, because information is lost in averaging. The square root of the JSD is a true metric (satisfying non-negativity, symmetry, identity of discernibles and triangle inequality), which is required by some clustering and manifold learning algorithms. The JSD can be viewed as defining a small neighbourhood around each cell, in which neighbours are near-indistinguishable in an information-theoretic sense. Exploiting this property, we construct an RNN graph, by imposing a fixed JSD radius around each cell, and connecting only cells that fall within this radius (for computational efficiency, we cap the maximum number of neighbours to a fixed number (usually 25)).

We clustered the RNN graph using modularity (Louvain) to define distinct sets of cells representing cell types or states along differentiation trajectories. We thus represent the high-dimensional gene expression manifold simultaneously as an RNN graph (facilitating analysis of continuous processes such as differentiation), a set of clusters (facilitating comparison between discrete cell states) and the components of the HPF (facilitating analysis of modular processes shared between cells).

For visualization, we computed HPF to 30 components and then used the Art of *t*-SNE heuristics[34] (but with exaggeration = 1.5, perplexity = 150) to obtain the *t*-SNE layout in Fig. 1. The use of a relatively small number of HPF components resulted in a globally smoother and more connected manifold, at the expense of reduced local resolution. This *t*-SNE was computed with commit 8a5518e81197f5aa9d54565cbce3875738d8dac9 of cytograph and used for visualization only.

For inference of longer trajectories along the manifold, we define a pseudolineage geodesic as the shortest path between two given cells (for example, a root cell and a terminally differentiated cell) on the RNN using the JSD as edge weights. The shortest path is therefore a path of small steps, each connecting a pair of near-identical cells. A pseudolineage tree of all cells can be constructed by designating a single root cell,

and computing the geodesics to all other cells. On this tree, many cells will be internal (branches), some external (leaves) and the root cell will have no predecessor. Cells not connected directly or indirectly to the root cell will not be part of the tree. For this paper, we constructed the pseudolineage tree by picking a single cell (no. 206,003) in the early neuroepithelium arbitrarily as the root.

Analysis of non-neuronal cells

Clustering. Cells belonging to a clade of the dendrogram containing glia (clusters 172–320) were pulled from the dataset. The dendrogram was cut into fifteen metaclusters that were subsequently annotated on the basis of gene expression. A new *t*-SNE was calculated on this subset of cells using cytograph with the parameters described in 'Cytograph 2.0', except using n_factors: 50 and mask: ['cellcycle', 'sex', 'ieg', 'mt', 'ery'] to mask cell-cycle, sex-related, immediate-early, mitochondrial and blood-related genes.

Clades containing fibroblasts (clusters 109–125) and immune cells (clusters 157–171) were also pulled from the dataset. Clusters were annotated on the basis of gene expression. A new *t*-SNE was calculated for each subset of cells using the parameters described in 'Cytograph 2.0' except that the mask: ['cellcycle', 'sex', 'ieg', 'mt', 'ery'], and n_factors: 50 parameters were used for fibroblast cells.

Differential expression. Differential expression was calculated using the diffxpy package. A *t*-test was calculated on log-normalized gene-expression values, in which each cell was normalized to the median total molecules per cell. Differentially expressed genes were then filtered for those with *q*-values less than 10^{-5} and expressed in at least 10% of either neurogenic radial glia or glioblasts.

Cell-type scores. Cell-type scores were based on the top 200 enriched genes for astrocytes and OPCs in the adolescent mouse dataset. The score was calculated as the percentage of total genes expressed in each cell that belonged to the enriched set.

Region classifier. One thousand two hundred cells were sampled from each of six subsets of the dataset: early radial glia (no. 196–223), late radial glia (no. 224–237 and 180–195), glioblasts (no. 282–311 and 172–179), OPCs (no. 312–318), neuroblasts (no. 321–491) and neurons (no. 492–797). Equal numbers of each subset were sampled from the forebrain, midbrain and hindbrain. Each subset was used to train a classifier for region that was then tested on the other five groups. The sklearn GradientBoostingClassifier was regularized with the parameter max_features, and max_features = 50 was chosen after testing the performance of the classifier over a range of values. Default values were used for other parameters.

Gastrulation and neural plate analysis

A previously published[17] dataset (https://github.com/MarioniLab/MouseGastrulationData) was prefiltered by removing cells not classified by the authors. Feature selection using a CV-mean was used to restrict the analysis to the most variable 3,500 genes. Analogously, 4,382 genes were selected for the E7–E8.5 dataset and the intersection between these two gene sets was then considered for the matching. The count table from the previous publication was imputed with *k*-nearest neighbour pooling ($k = 30$) and then log-transformed. To place the E7–E8.5 cells belonging to the first 4 cluster clades (clusters no. 0–no. 14, no. 15–no. 36, no. 37–no. 68, no. 69–no. 86) on the *t*-SNE embedding for ref. [17], we first computed the similarity (Pearson's *R*) between each of the E7–E8.5 cells and the reference cells. Then, we assigned to each cell the median of the embedding coordinates computed over the 20 most similar cells in the reference.

For an in-depth analysis of the early neuroepithelium, E7–E8.5 cells of clusters no. 51–55 were further analysed by running cytograph with the parameters described in 'Cytograph 2.0', resulting in the ten subclusters described in Extended Data Fig. 4. For the early neuroblast analysis, we queried clusters of the class 'neuroblast' for cells sampled at ages younger than E9.0.

In situ sequencing analysis

Channel images were merged into a multichannel .tiff image and aligned to a reference round. The channel used for tile alignment was the DAPI stain. Following alignment, the tiles were stitched together using Microscopy Image Stitching Tool[35] and subsequently split into non-overlapping 2,000-by-2,000-pixel tiles. Each individual tile was then top-hat filtered, the rolling circle products (RCPs) were segmented and the intensity was measured in each channel. The RCP was then assigned to the base with the highest intensity. The public repository for the code used can be found at https://github.com/Moldia/Tools. Gene expression was summarized considering an isotropic grid with step 16 μm and counting the called dots for each gene. The data were smoothed with Gaussian filter with a bandwidth of 24 μm. Grid locations with a total of fewer than 4 called dots were excluded from the clustering analysis. As a preprocessing before clustering, the counts were transformed with the variance stabilizing transform $\log_2(x+1)$, a principal components analysis was computed and the top 40 principal components were retained. Finally, we ran a *k*-means clustering, setting n_cluster = 40.

Generalized Tangram (Bone Fight)

Tangram[28] was designed to map single-cell data to spatial gene-expression data, computing a projection matrix of shape n_cells × n_voxels using gradient descent on a loss function that combines gene-expression correlation and a density term. However, our dataset comprises 292,495 single cells and 750,000 in situ sequencing voxels, which means that the projection matrix would occupy almost one terabyte of RAM (assuming 32-bit floats). Optimizing such a matrix would be intractable on any reasonable hardware. However, our goal here was not to map individual cells, but only to obtain a spatial distribution for each of the 798 clusters. We therefore generalized Tangram to accept a volume prior on the source dataset, which sets the expected spatial volume of each entity in the dataset. For single-cell data, a uniform prior yields results near-identical with standard Tangram. For cluster-level data, the prior volume can be set to the number of cells per cluster, thus ensuring that large clusters are allowed to occupy a large spatial volume. After this modification, the projection matrix for our datasets was only 2.4 gigabytes and the optimization was orders of magnitude faster, converging in a few minutes on a high-end laptop.

Reporting summary

Further information on research design is available in the Nature Research Reporting Summary linked to this paper.

Data availability

RNA-seq data are available at the Sequence Read Archive (https://www.ncbi.nlm.nih.gov/sra) under accession PRJNA637987. HybISS data are available at http://mousebrain.org/downloads. Single-cell gastrulation data were downloaded from https://github.com/MarioniLab/MouseGastrulationData.

Code availability

Source code is available at https://github.com/linnarsson-lab.

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Author contributions S.L. supervised the project. S.L. and G.L.M. conceived the study design. G.L.M., K.S. and S.L. analysed, annotated and interpreted the single-cell data and wrote the manuscript. G.L.M. and A.F. performed the single-cell experiments. A.F. dissociated embryos. A.J. prepared sequencing libraries. P.L. and S.L. built the companion website. E.V., D.G. and C.M.L. performed in situ sequencing experiments. A.M.A. performed and analysed smFISH experiments. E.V. prepared tissue sections and performed anatomical annotation. K.S., I.K., A.R.L., G.L.M. and C.M.L. analysed and interpreted in situ sequencing data. L.M.D. performed clustering sensitivity analysis. M.N. supervised in situ sequencing experiments. All authors critically reviewed the manuscript and approved the final version.

Competing interests M.N. serves as a scientific advisor for 10x Genomics. All other authors declare no competing financial interests.

Additional information

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Correspondence and requests for materials should be addressed to G.L.M. or S.L.

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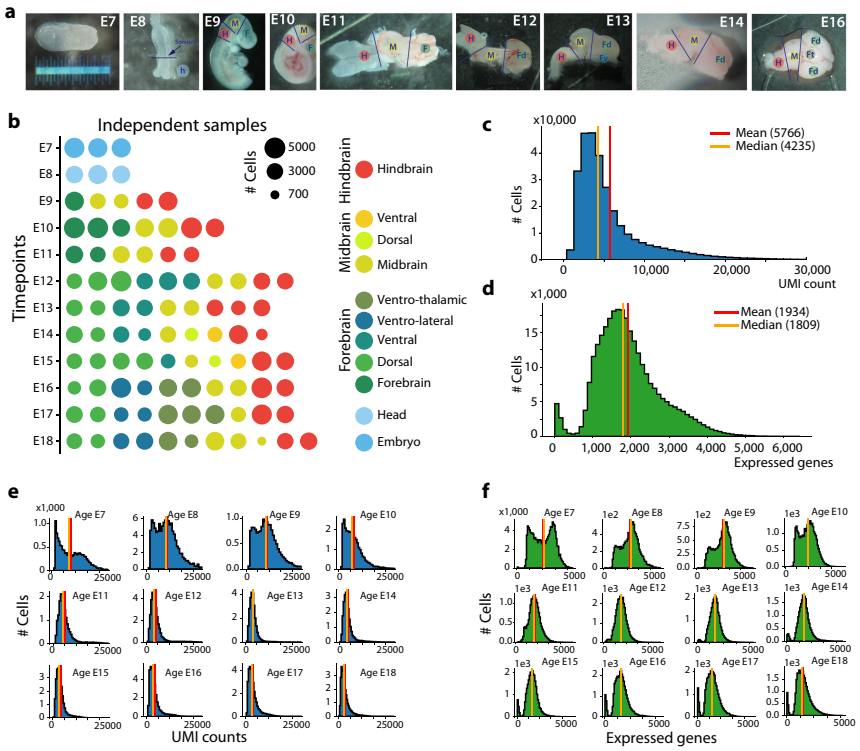
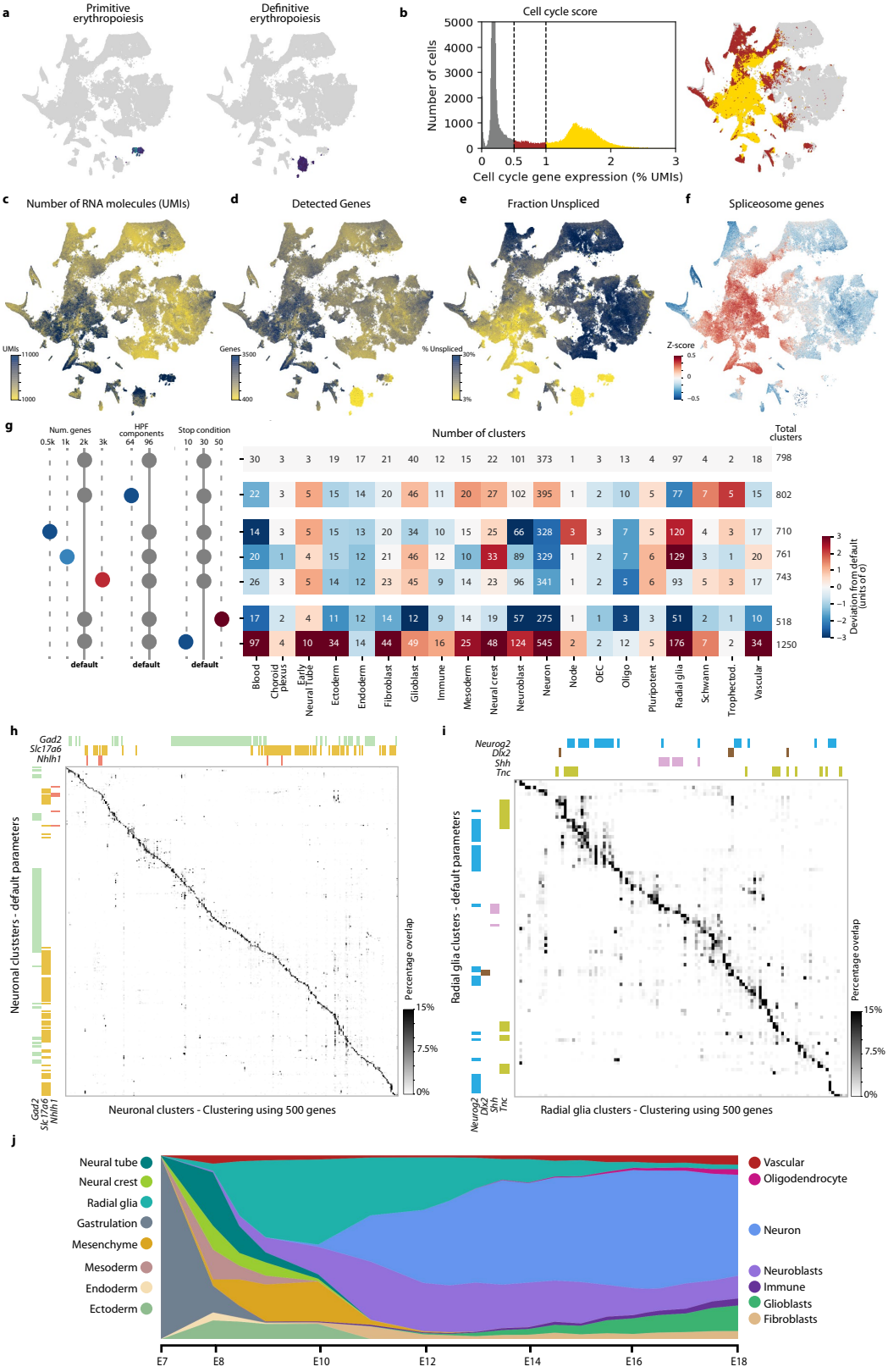
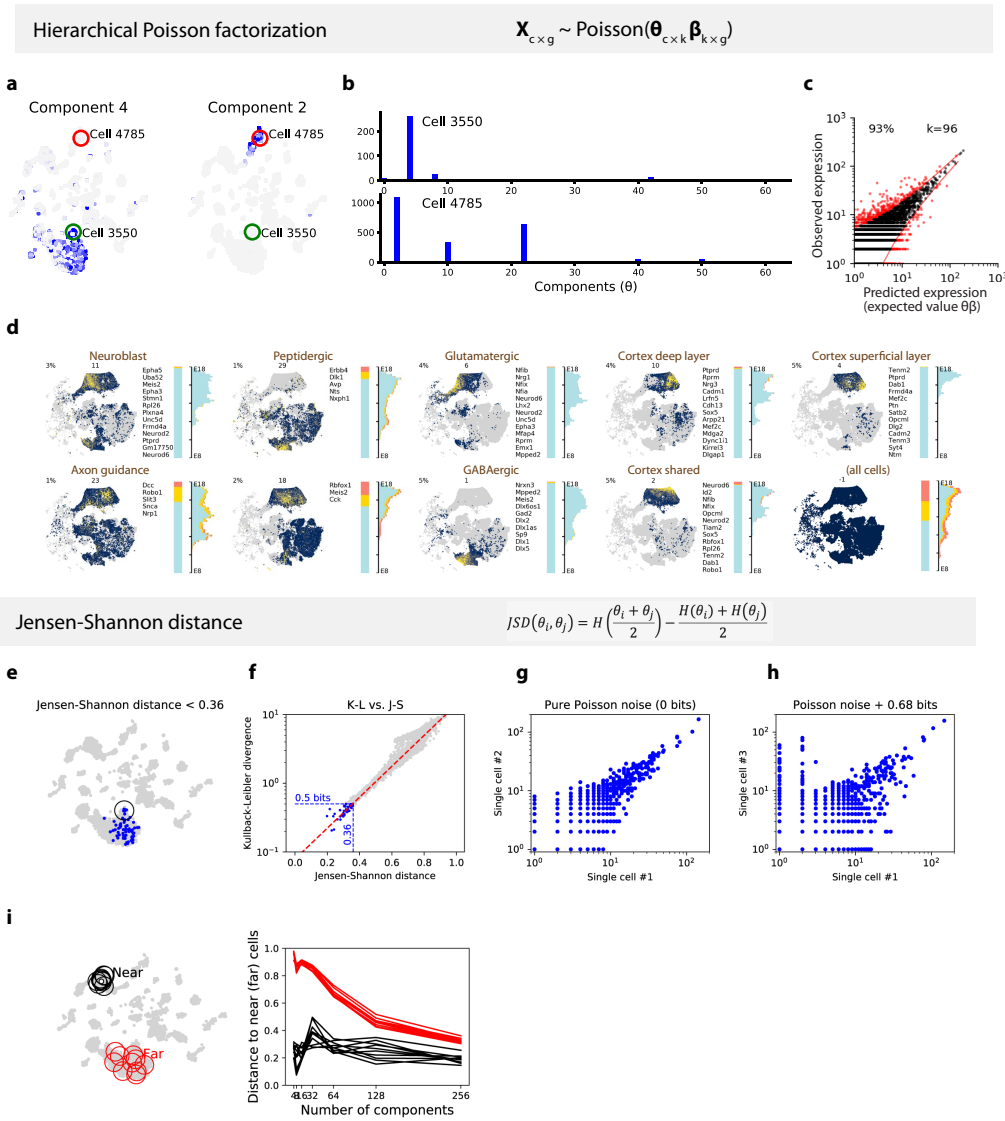


Extended Data Fig. 1 | Experimental design and data quality. a, Stereo microscope photographs illustrating the tissue dissection strategy. **b**, Samples composing the dataset. The area of each circle is proportional to the number of cells sampled. Colour indicates the region dissected. **c**, Histogram showing the distribution of number of UMIs per cell detected across the entire dataset. **d**, Histogram showing the distribution of number of genes detected per cell across the entire dataset. **e**, Distribution of UMIs per cell aggregated per age group. **f**, Distribution of number of genes detected per cell aggregated per age group.

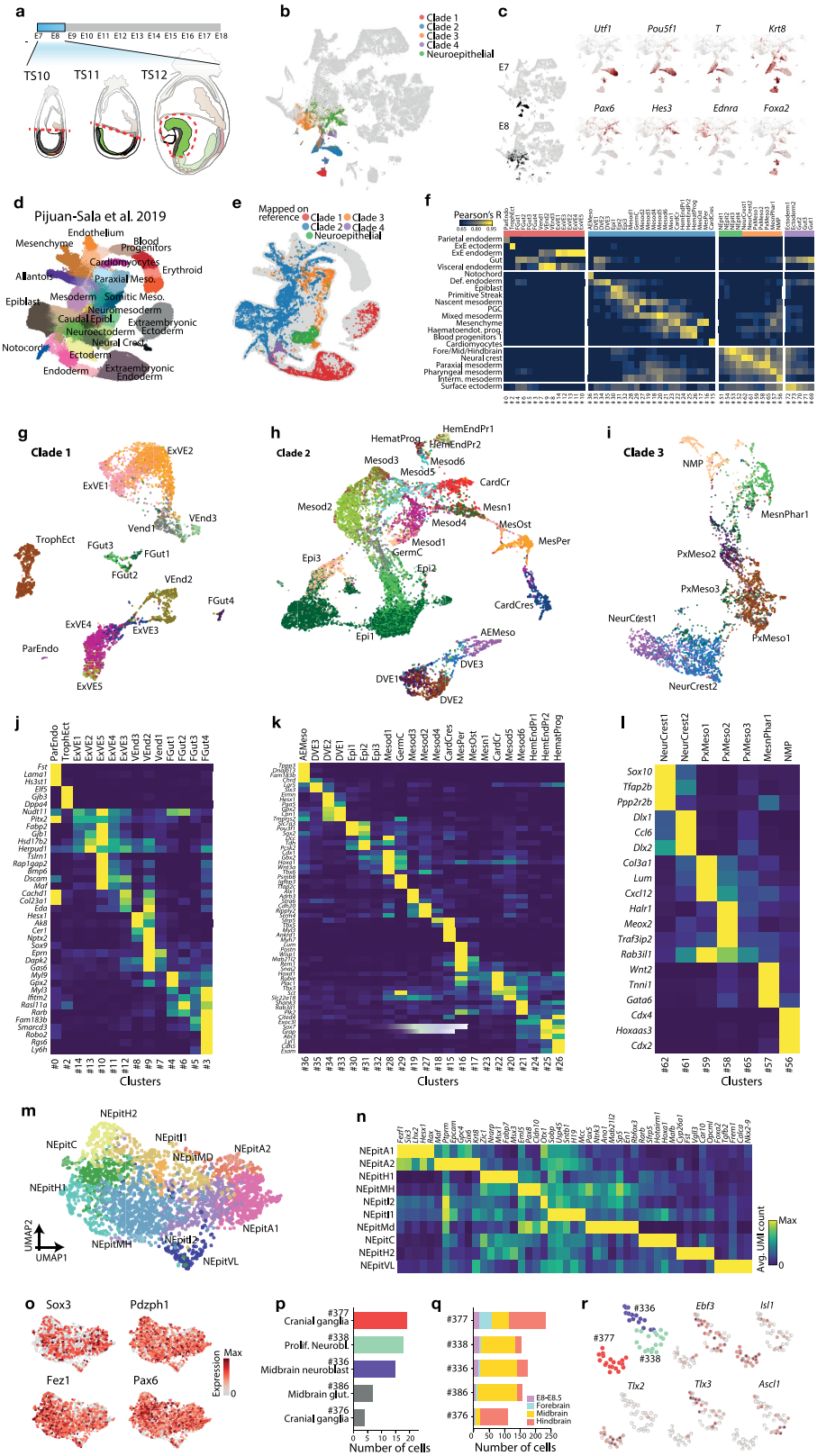


Extended Data Fig. 2 | See next page for caption.

Extended Data Fig. 2 | Global properties of gene expression during brain development. a, Erythrocyte clusters expressing *Hbb-bh1* (>500 UMIs per cell) (primitive erythropoiesis) or *Hbb-bs* (>5,000 UMIs per cell) (definitive erythropoiesis). **b**, Histogram of cell cycle scores (left) and cell cycle scores indicated on the main *t*-SNE (right). **c**, Distribution of total UMIs per cell. **d**, Distribution of total number of genes detected per cell. **e**, Distribution of the fraction of unspliced reads detected per cell. **f**, Average *z* score for a set of genes with GO term related to the spliceosome. **g**, Number of clusters observed for each major cell class, as a function of varying hyper-parameters of the clustering algorithm (indicated, left). Top row shows the default clustering. Each subsequent row varies one or more hyper-parameters (number of genes selected, number of HPF components and the stop condition), resulting in different numbers of clusters per class. Deviations from the default settings are indicated by the colours. **h**, Confusion matrix comparing clustering results for neurons using default settings (vertical axis) and clustering using only 500 genes (horizontal axis). **i**, Same as **h**, but for radial glia. The expression of key marker genes is indicated along the axes to reveal major subclasses. **j**, Distribution of major classes of cells by gestational age.

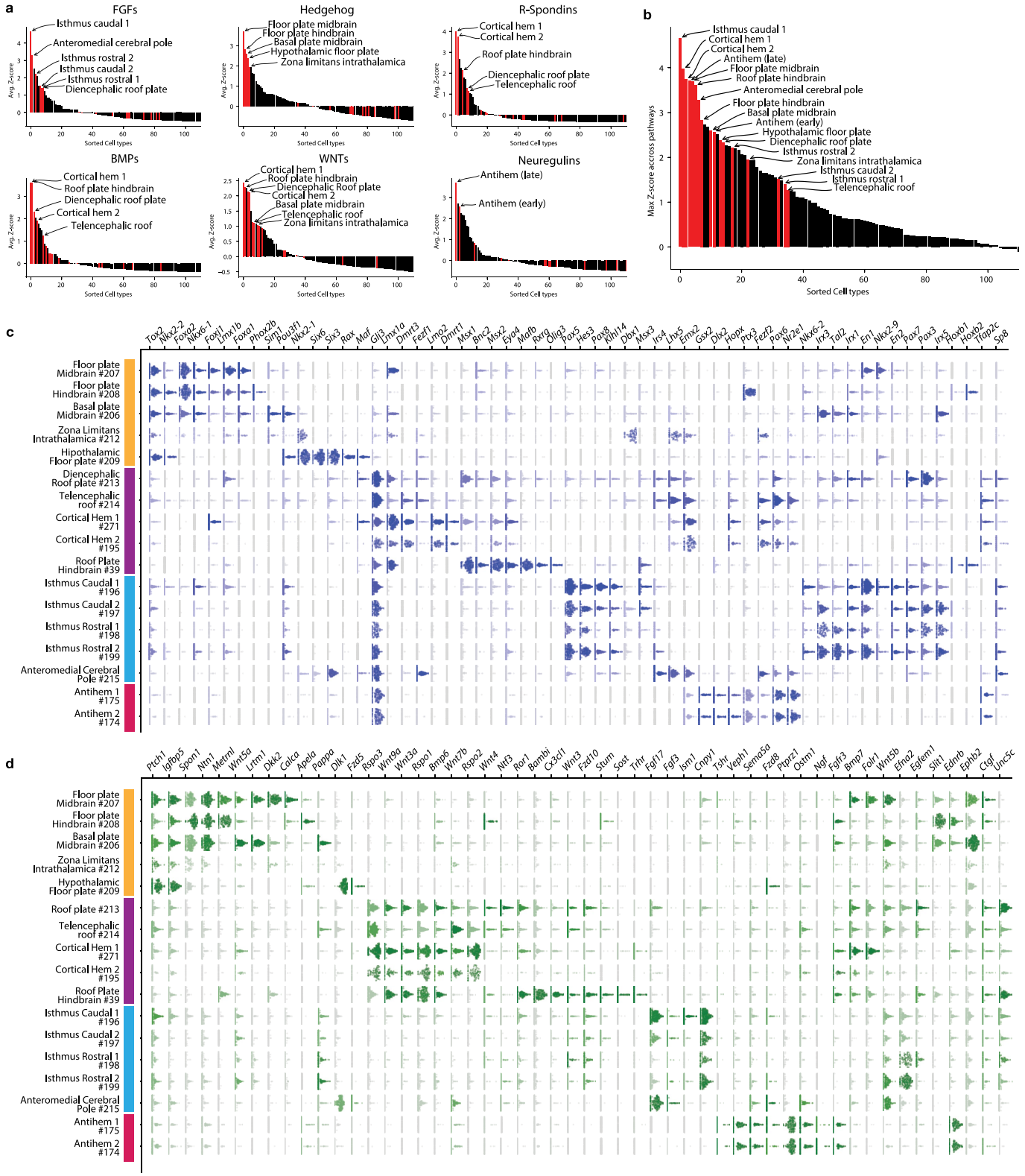


Extended Data Fig. 3 | Computational methods. a, Illustration of HPF on a small subset of cells using 32 components, showing component loadings across cells for components 4 and 2, illustrating the modularity of components. **b**, Loadings of all components for two single cells, illustrating the sparseness of components; each cell is approximated by a small number of non-negative components. **c**, Scatter plot showing observed expression of all genes in all cells, versus predicted expression based on 96 HPF components. Ninety-three per cent of data points were located inside the 95% confidence intervals of the Poisson distribution, demonstrating the accuracy of the HPF representation. **d**, HPF analysis of all neuronal cells from the full dataset using 32 components, showing that HPF captures essential features of forebrain neurogenesis. Each subplot shows a *t*-SNE coloured by component loading, a list of genes with high loading in the component, and a bar chart and histogram showing the regional distributions of cells. Nine selected components are shown, related to forebrain based on the regional distribution of cells with high loadings. **e**, Example of cells (blue) that fall within a Jensen–Shannon distance of 0.36 of a single cell (black circle), corresponding to 0.5 bits of Shannon entropy. **f**, The relationship between the Kullback–Leibler divergence and the Jensen–Shannon distance, showing that in this case 0.36 JSD corresponds to 0.5 bits. **g**, Two virtual cells (Poisson samples drawn from a single HPF vector), showing the distribution of pure Poisson expression noise. **h**, The effect of switching two HPF components, resulting in a change in expression of many genes and a Jensen–Shannon distance of 0.68 bits. **i**, The 'curse' of dimensionality. As the number of HPF components is increased, the difference in Jensen–Shannon distance between nearest-neighbours and far-away cells on the manifold decreases.

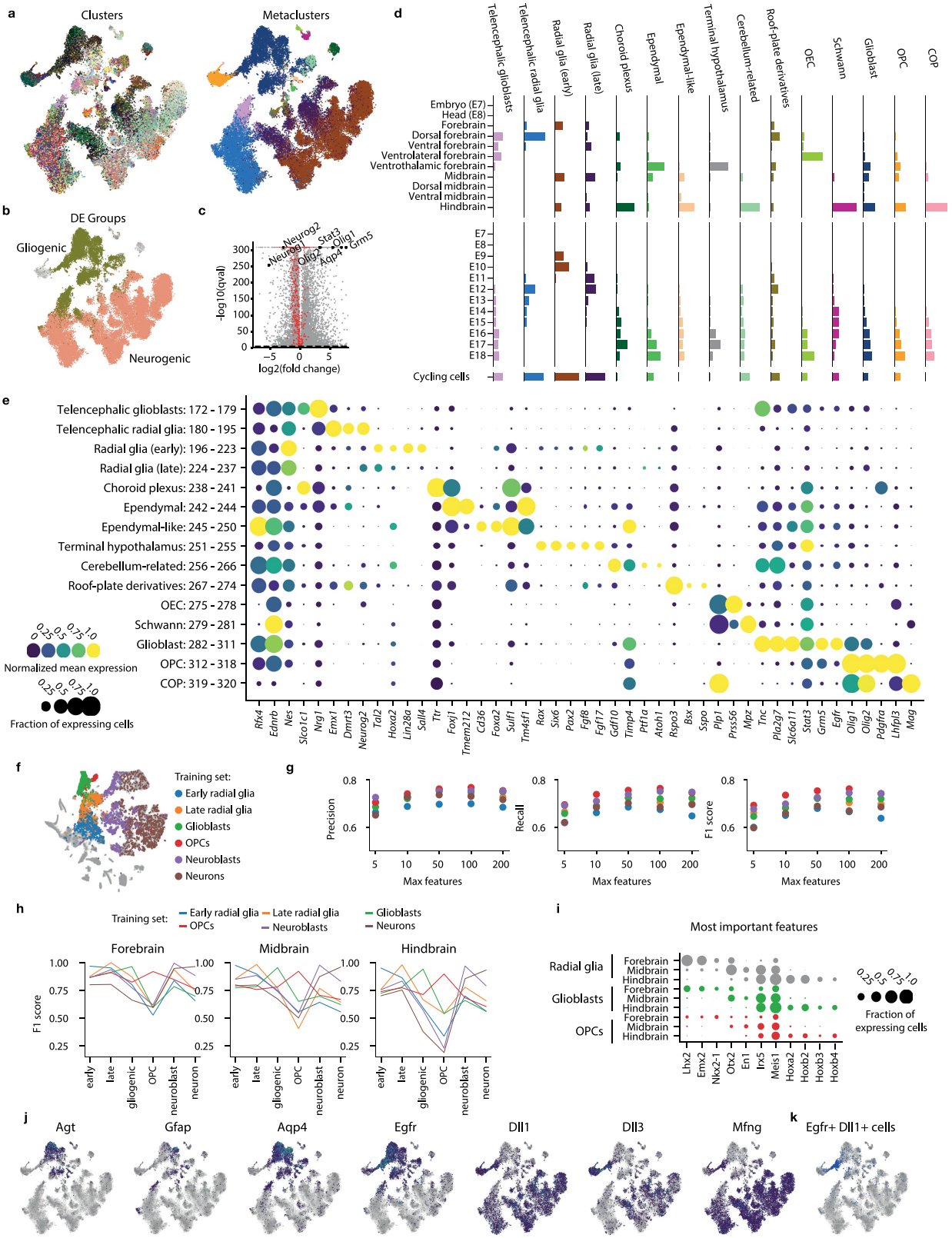


Extended Data Fig. 4 | See next page for caption.

Extended Data Fig. 4 | Emergence of cell-type heterogeneity from gastrulation to neurulation. a, Scheme of tissue sampling between E7 and E8.5. **b**, *t*-SNE from Fig. 1 overlaid with cells sampled E7–E8.5 and belonging to dendrogram clades 1–4. **c**, *t*-SNE overlaid with expression of lineage-specific marker genes. **d**, Uniform manifold approximation and projection (UMAP) of the mouse gastrulation single-cell atlas from ref. [17]. **e**, Cells from **b** projected onto the embedding in **d**[17]. **f**, Pairwise correlation between neurulation-stage clusters of this Article and those of ref. [17]. **g**–**i**, UMAP embedding of cells of the first (**g**), second (**h**) and third (**i**) cluster clade of the dendrogram. Cells are coloured and labelled by cluster identity. **J**–**l**, Heat map showing genes enriched in of the first (**j**), second (**k**) and third (**l**) cluster clade. **m**, UMAP embedding of subclustered neuroepithelial cells obtained from time points E7–E8.5. Cells are coloured by cluster. **n**, Heat map displaying expression of genes enriched in each of the neuroepithelial populations. **o**, Expression of broad neuroepithelial markers overlaid on the UMAP in **m**. **p**, Bar plot reporting the number of E7–E8.5 cells that clustered with the indicated neuroblast clusters. Clusters are sorted by number of cells. Bars are coloured as in **l**. **q**, Bar plots showing the dissection origins of the cells belonging to the clusters in **p**. **r**, UMAP representation of the E8.5 cells belonging to clusters no. 377, no. 366 and no. 338 (top left), and overlaid with expression of enriched transcription factors.

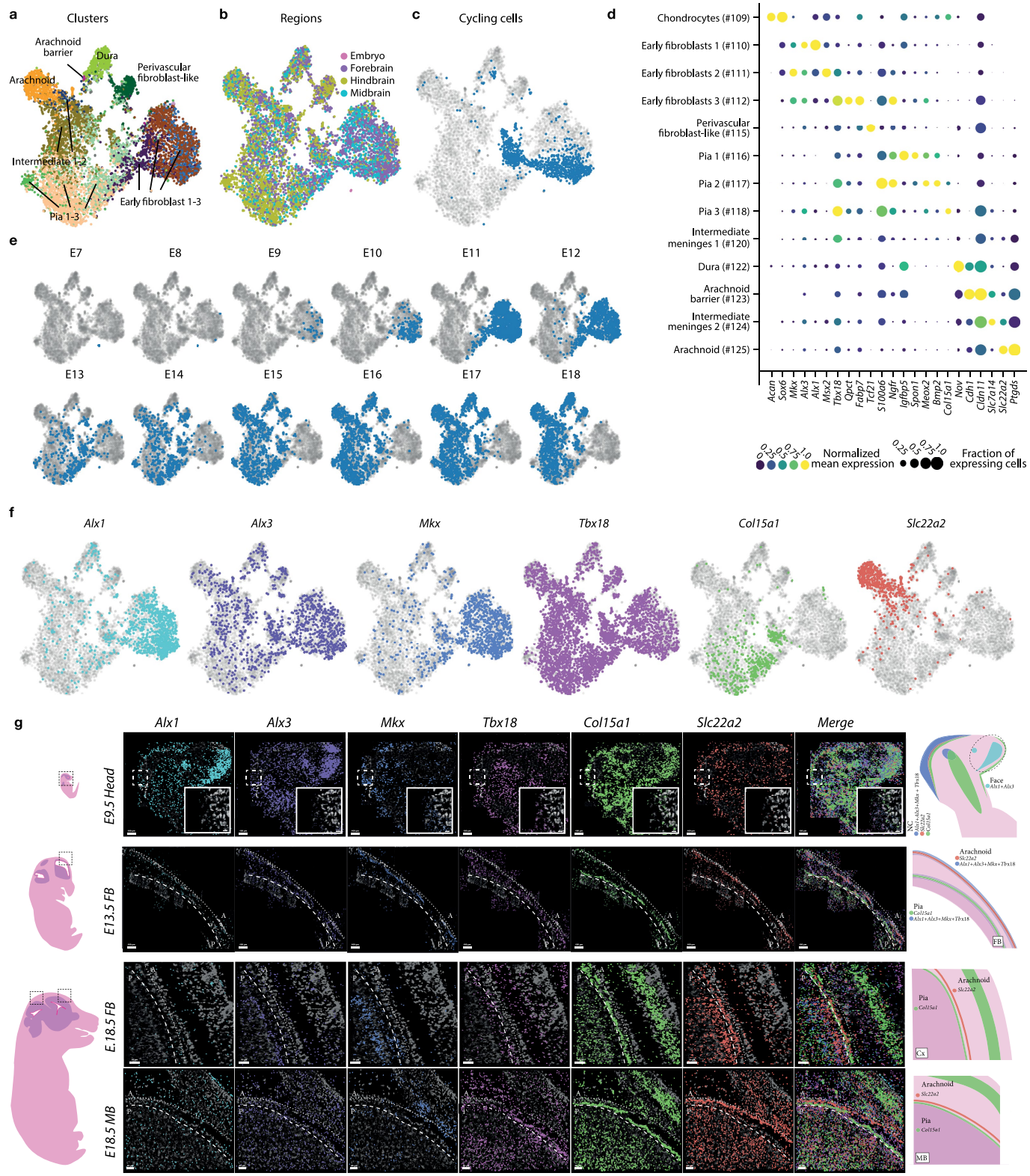


Extended Data Fig. 5 | Characterization of molecular identity and signal-receptor repertoire of neural tube organizers. a, Cell types sorted on the basis of the average *Z* score of morphogens genes grouped in categories on the basis of gene family and neurulation literature. Hedgehog, *Shh* and *Ptch1*; R-spondins, *Rspo1*, *Rspo2*, *Rspo3* and *Rspo4*; Wnts, *Wnt1*, *Wnt3a*, *Wnt5a* and *Wnt8b*; neuregulins, *Nrg1*, *Nrg3* and *Fgf7*; Fgfs, *Fgf8*, *Fgf15*, *Fgf17* and *Fgf18*; Bmps, *Bmp6*, *Bmp7* and *Gdf7*. Cell types displayed in red were localized and analysed in Fig. 2 and in the panels below. **b**, Summary of the plots in **a** showing the maximum average *z* score achieved. **c**, Beeswarm plots showing the expression of a selection of transcription factors that were found to be enriched in the different organizer cell populations. **d**, Beeswarm plots showing the gene expression of enriched receptors and ligands with a distribution in relation to organizers that, to our knowledge, had not previously been described. Genes in **c**, **d** were selected for HybISS profiling (Fig. 4).

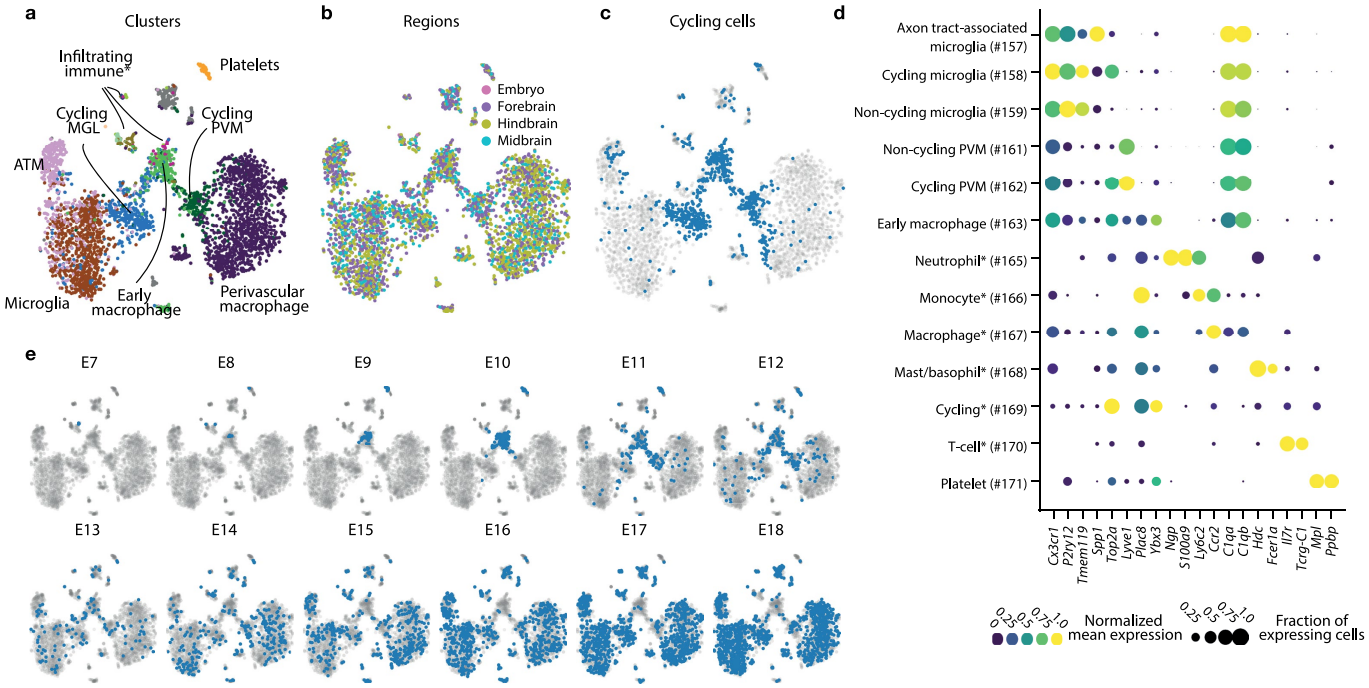


Extended Data Fig. 6 | See next page for caption.

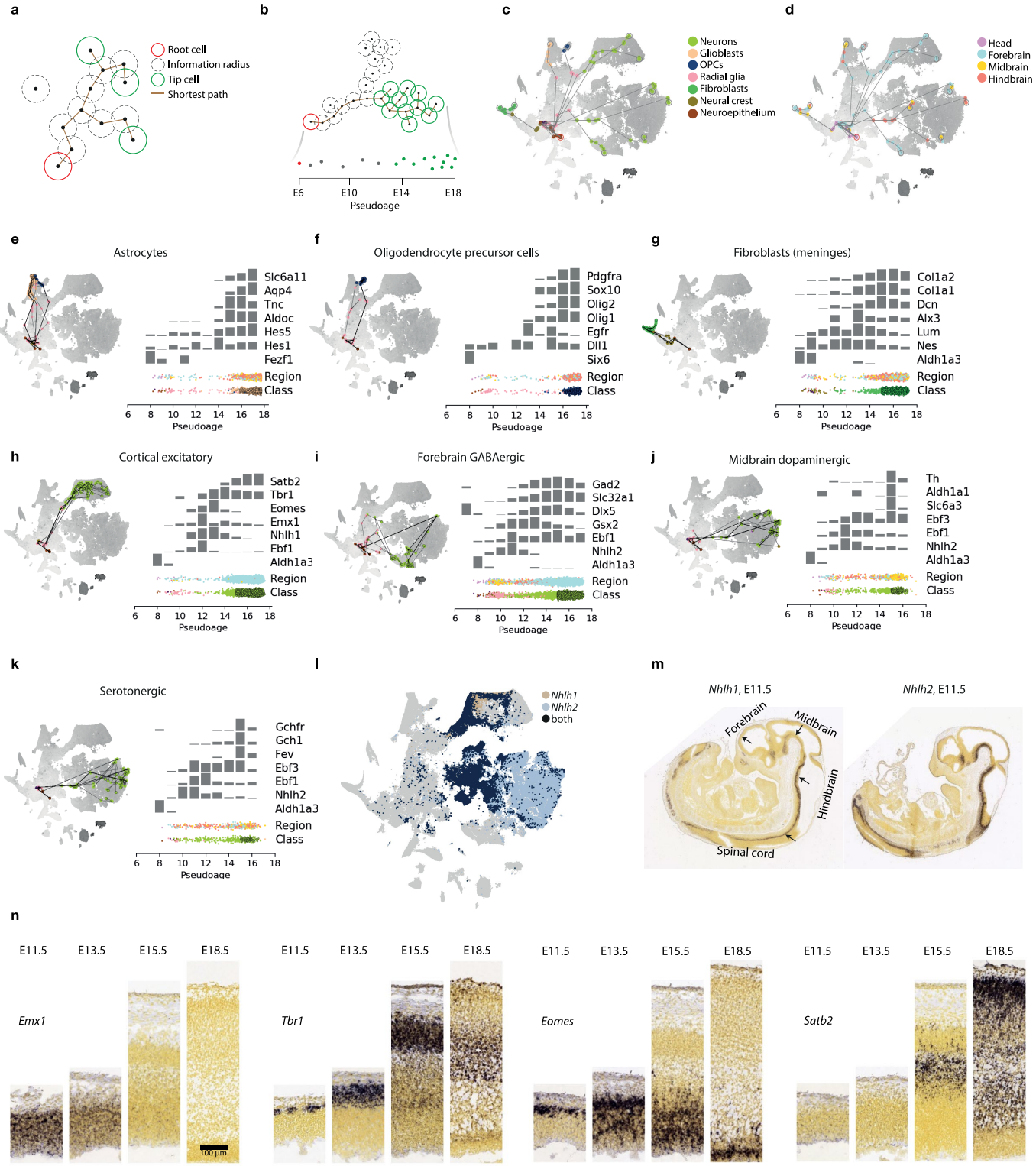
Extended Data Fig. 6 | Glial diversity. a, Cells from clusters 172 to 320 were pulled from the complete dataset, and a *t*-SNE embedding was calculated. The dendrogram was cut into 15 metaclusters. Cells are coloured by their original cluster (left) and metacluster cut from the dendrogram (right). **b**, Cells are coloured by their labels for differential gene-expression testing: brown, glioblast; coral, neurogenic; grey, not included in testing. **c**, Select genes are annotated on a volcano plot illustrating differential expression between neurogenic and gliogenic radial glia. The dotted line denotes the significance threshold ($q = 10^{-5}$). The range of the *x* axis was chosen to capture statistically significant genes. Cell-cycle genes are coloured red. **d**, For each metacluster, the size of the bar indicates the percentage of cells from each tissue (top), each embryonic stage (middle) or that are cycling (>1% cell-cycle UMIs) (bottom). **e**, Expression dot plots are shown for select genes. The size of the dot represents the fraction of cells expressing a gene. The colour represents normalized mean expression, in which the mean expression for each gene is normalized by the maximum mean expression of the gene across the metaclusters. **f**, Cells selected for region classification are coloured on the *t*-SNE. Grey cells were not selected. **g**, For a range of parameter values (max_features), a gradient boosting classifier was fit to each training set and tested on the five remaining training sets, coloured as in **f**. Average precision, recall and F1 scores are plotted for each parameter value and training set. **h**, F1 scores for each gradient boosting classifier (max_features = 50) are shown for forebrain, midbrain and hindbrain cells. The colour of each line indicates which cells were used to train the classifier (coloured as in **g**). **i**, Feature importance for each gradient boosting classifier was used to rank genes, and the intersection of the 100 most important genes for all 6 classifiers is shown on the *x* axis. Expression dot plots are shown for each of these genes in radial glia, glioblasts and OPCs from the forebrain, midbrain and hindbrain . **j**, Cells are coloured by log-transformed gene expression for the indicated genes. Grey indicates no expression. **k**, Cells that are positive for both *Egfr* and *Dll1* are coloured blue.



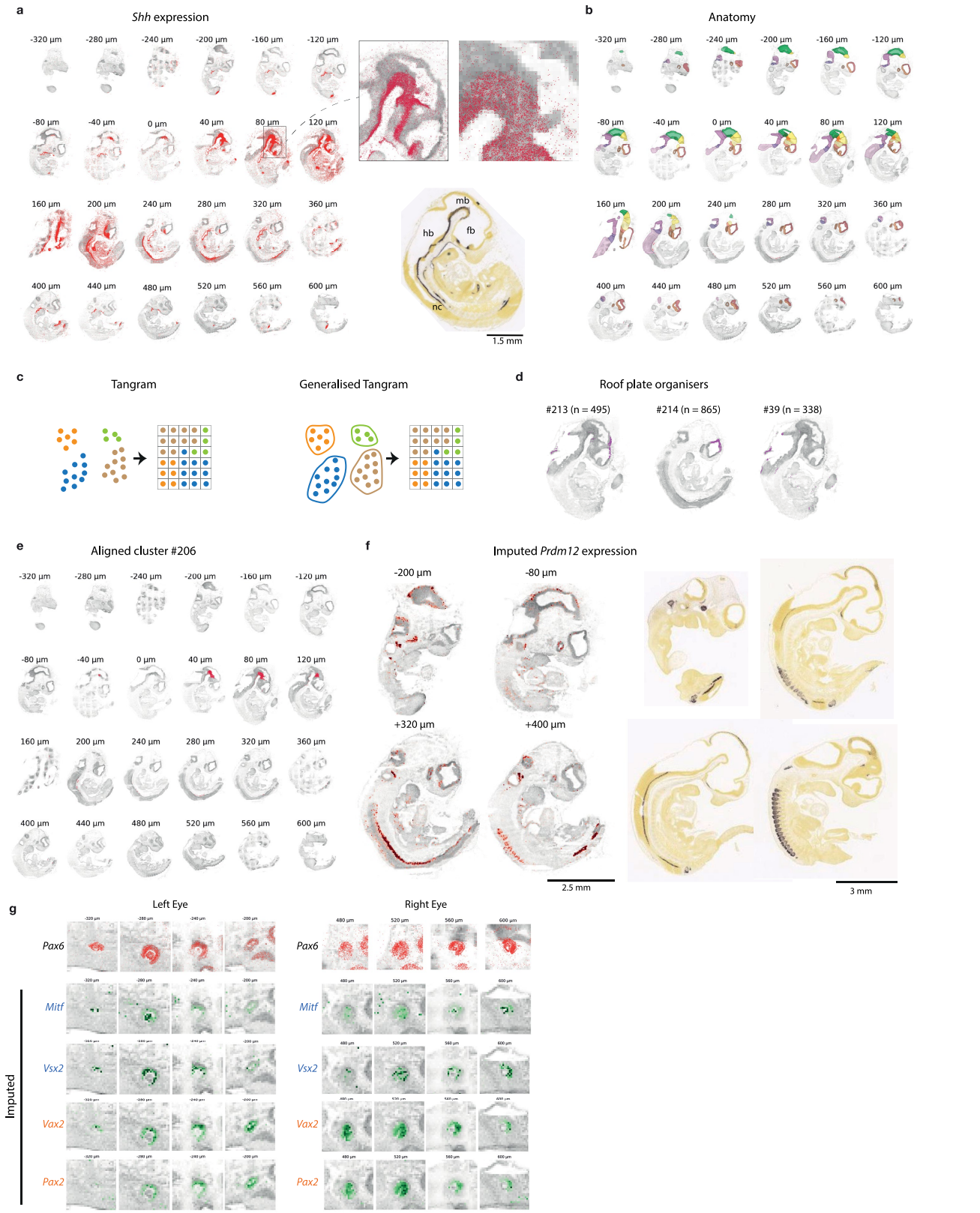
Extended Data Fig. 7 | Fibroblasts. Cells from clusters 109 to 125 were pulled from the dataset, and a *t*-SNE embedding was calculated. **a**, Cells are coloured by their clusters. Putative doublets are grey. **b**, Cells are coloured by their tissues of origin. **c**, Cycling cells (>1% cell-cycle UMIs) are coloured blue. **d**, Expression dot plots are shown for select genes. **e**, Cells from the indicated embryonic age are coloured blue on each *t*-SNE. **f**, Cells are coloured if they express the indicated gene. **g**, Multiplexed single-molecule RNA fluorescence in situ hybridization for six marker genes (columns) at three different ages (rows). Cartoons on the left indicate the approximate location of each image. Cartoons on the right show an interpretation of the observed gene expression patterns. This experiment was performed once.



Extended Data Fig. 8 | Immune cells. Cells from clusters 157 to 171 were pulled from the dataset, and a *t*-SNE embedding was calculated. **a**, Cells coloured by their clusters. Putative doublets are grey. Infiltrating immune includes six clusters, starred in **d**. ATM, axon tract-associated microglia; MGL, microglia; PVM, perivascular macrophage. **b**, Cells coloured by their tissues of origin. **c**, Cycling cells (>1% cell-cycle UMIs) coloured blue. **d**, Expression dot plots shown for select genes. **e**, Cells from the indicated gestational age coloured blue on each *t*-SNE.



Extended Data Fig. 9 | Pseudolineage analysis and neurogenesis. **a**, Pseudolineage tree algorithm, computing shortest paths to the root cell passing only through neighbours inside the information radius (that is, neighbours within a fixed maximum Jensen–Shannon distance). **b**, Isolating the pseudolineages terminating in a selected cluster, and projecting to the pseudoage axis. **c**, Ten selected pseudolineages coloured by major class, on the *t*-SNE of Fig. 1 with grey scale showing the geodesic distance to the root from every cell. **d**, The same ten selected pseudolineages coloured by tissue. **e**–**i**, Pseudolineages of astrocytes (**e**), OPCs (**f**), fibroblasts (**g**), cortical excitatory neurons (**h**), forebrain GABAergic neurons (**i**), midbrain dopaminergic neurons (**j**) and hindbrain serotonergic neurons (**k**). Each plot shows a randomly selected subset of pseudolineages terminating in the indicated clusters, as well as expression of selected genes in pseudoage bins along the lineage (calculated for all cells in the lineage). The region and class of each cell is indicated at the bottom. **l**, Expression of *Nhlh1* and *Nhlh2*. **m**, In situ hybridization for *Nhlh1* and *Nhlh2* on sagittal sections of the E115 mouse embryo (Allen Brain Atlas). **n**, RNA in situ hybridization of mouse embryonic brain at the indicated time points, showing genes relevant to the cortical lineage (image credit: Allen Institute). Each subpanel shows a strip from ventricular zone to pia at four different ages.



Extended Data Fig. 10 | Spatial mapping of single-cell clusters. a, Expression of *Shh* detected by in situ sequencing on all sections, with close up images showing the detection of individual mRNA molecules. Inset shows the expression of *Shh* by in situ hybridization (Allen Brain Atlas) for reference. **b**, Manually curated anatomical annotation of all tissue sections (coloured as in the Allen Brain Atlas). **c**, Illustration of Tangram (mapping single cells to spatial voxels) and our generalized Tangram (mapping clusters to voxels). **d**, Inferred spatial distribution of roof plate organizers (clusters no. 213, no. 214 and no. 39). **e**, Inferred spatial distribution of cluster no. 206 on all tissue sections. **f**, Imputed expression of *Prdm12*, shown for four informative sections (left) and Allen Brain Atlas ground truth (right). **g**, Spatial gene expression of *Pax6* and imputed gene expression of dorsal (*Mitf* and *Vsx2*) and ventral (*Vax2* and *Pax2*) optic vesicle markers in the left and right eye.

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Corresponding author(s): Sten Linnarsson

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☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
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☒	☐	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used during data collection, other than as part of Illumina and 10x Genomics instruments.
Data analysis	All custom source code is available at https://github.com/linnarsson-lab (repositories cytograph2, auto-annotation-md, BoneFight). We used cellranger v3 (10x Genomics) and velocyto 0.17.17 for primary analysis of single-cell sequencing data.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

RNA-seq data is available at the Sequence Read Archive (SRA; https://www.ncbi.nlm.nih.gov/sra) under accession PRJNA637987. HybISS data is available at http://mousebrain.org/downloads.

Field-specific reporting

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The number of animals and the number of cells analyzed per animal was determined by previous experience (particularly our adolescent cell atlas, Zeisel et al. Cell 2018). An average of five animals were used per age and tissue (minimum 2). Detailed descriptions of all the samples are given in Table S1.
Data exclusions	Some samples were excluded due to failures of the 10x controller (leading to broken emulsion). Experiments with broken emulsions were considered failed, did not result in any data and were excluded from this study. All remaining experiments are listed in Table S1 and were included in the study. However, specific cells were excluded based on low UMI count (< 2000), and specific clusters were excluded based on high average doublet score (>0.3). 20 additional clusters were removed after manual annotation as doublets or contamination based on their gene expression. Extensive quality controls were performed on all samples as described in the methods section, which led to the exclusion of some samples and cells.
Replication	Due to the size and cost of the experiment, full replication was not feasible. We instead validated our findings against previous work and by using in situ hybridization to validate observations.
Randomization	The study is descriptive and did not report contrasts between groups, hence randomization was not applicable.
Blinding	Blinding was not applicable since no hypothesis was tested (the study is descriptive). All of our data originated from wild-type embryonic mouse brain. The purpose of our work was to describe the molecular heterogeneity of the wild-type brain. There were therefore no conditions we were able to blind.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Animals and other organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research

Laboratory animals	We used pregnant female CD-1 mice and collected embryos at the ages indicated in Table S1
Wild animals	No wild animals were used
Field-collected samples	No field samples were used
Ethics oversight	All animal procedures were approved by the Stockholm ethics committee (N68/14; Stockholms djurförsöksetiska nämnd) and followed Directive 2010/63/EU of the European Parliament and of the Council, the Swedish Animal Welfare Act (Djurskyddslagen: SFS 1988:534), the Swedish Animal Welfare Ordinance (Djurskyddsförordningen: SFS 1988:539) and the provisions regarding the use of animals for scientific purposes: DFS 2004:15 and SJVFS 2012:26.

Note that full information on the approval of the study protocol must also be provided in the manuscript.